



Contents of This Lecture

5.1 Introduction

5.2 Frequency Response of Systems

5.3 Distortionless Transmission

5.4 Ideal Low-Pass Filter

5.5 Physical Realizability of Systems

5.6 Modulation and Demodulation

5.7 Recovering Continuous-Time Signals from Sampled Signals

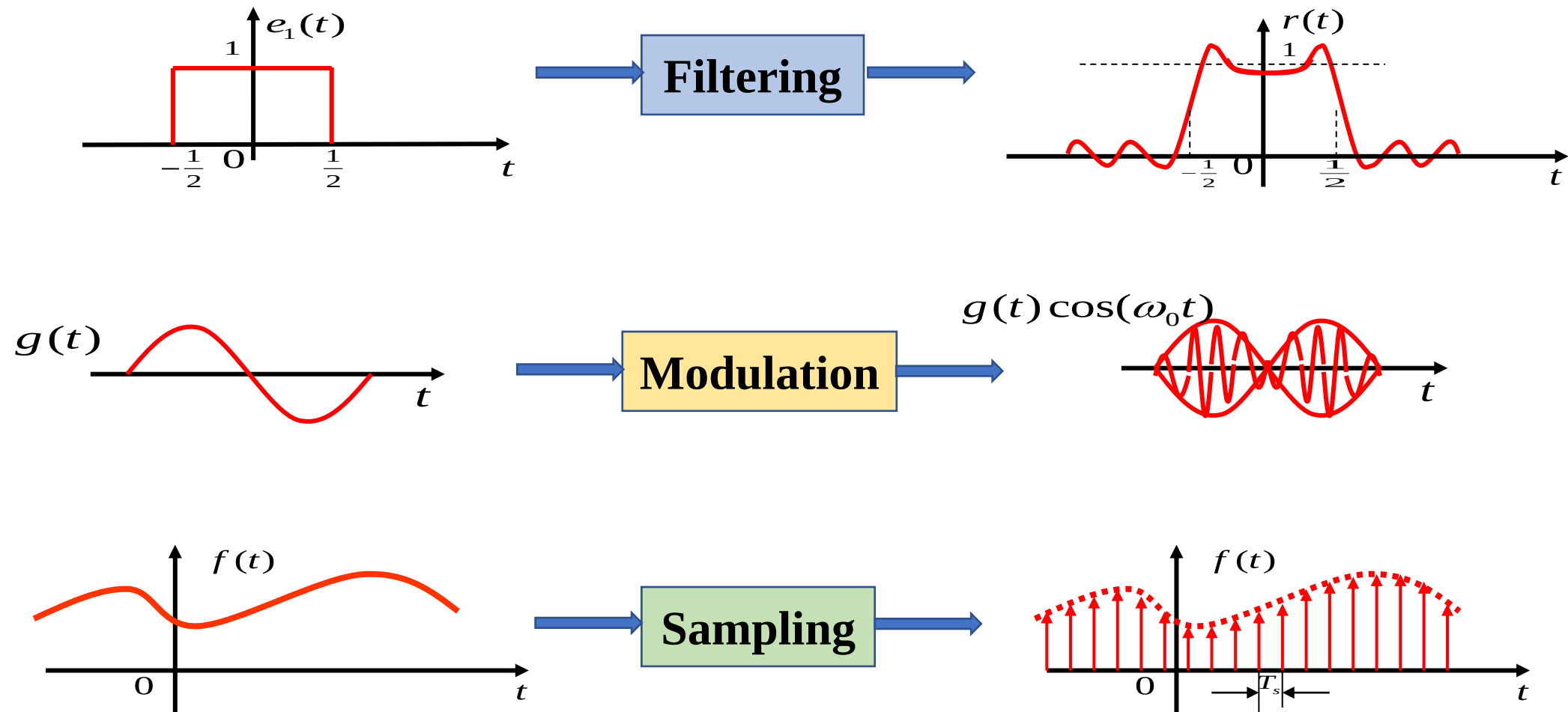
5.8 Frequency-Division Multiplexing (FDM) and Time-Division Multiplexing (TDM)

Learning Objectives of This Lecture

1. Understand the **physical meaning** of the frequency-domain system function;
2. Familiarize with **the time-domain and frequency-domain conditions for distortionless transmission**;
3. Familiarize with the **frequency-domain design principles** of mathematically **ideal filters** such as ideal low-pass filters;
4. Use time-domain **causality** and the frequency-domain **Paley-Wiener criterion** to judge the **physical realizability** of systems;
5. Master the frequency-domain analysis of **modulation** and **demodulation**;
6. Understand the **holding** and frequency-domain analysis of sampled signals;
7. Understand **multiplexing**.

5.1 Introduction

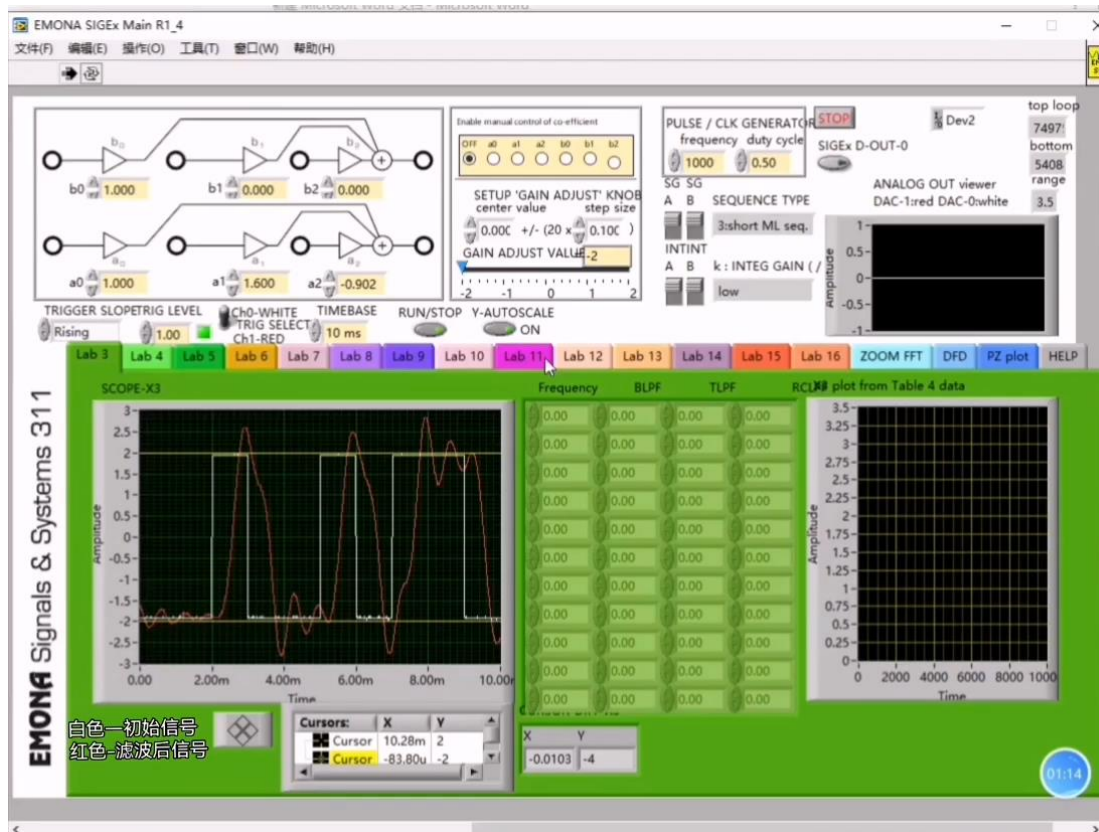
- **Frequency-domain analysis in communication systems** reveals signals' inherent frequency characteristics and the close relationship between their time and frequency characteristics, deriving key concepts such as spectrum and bandwidth.



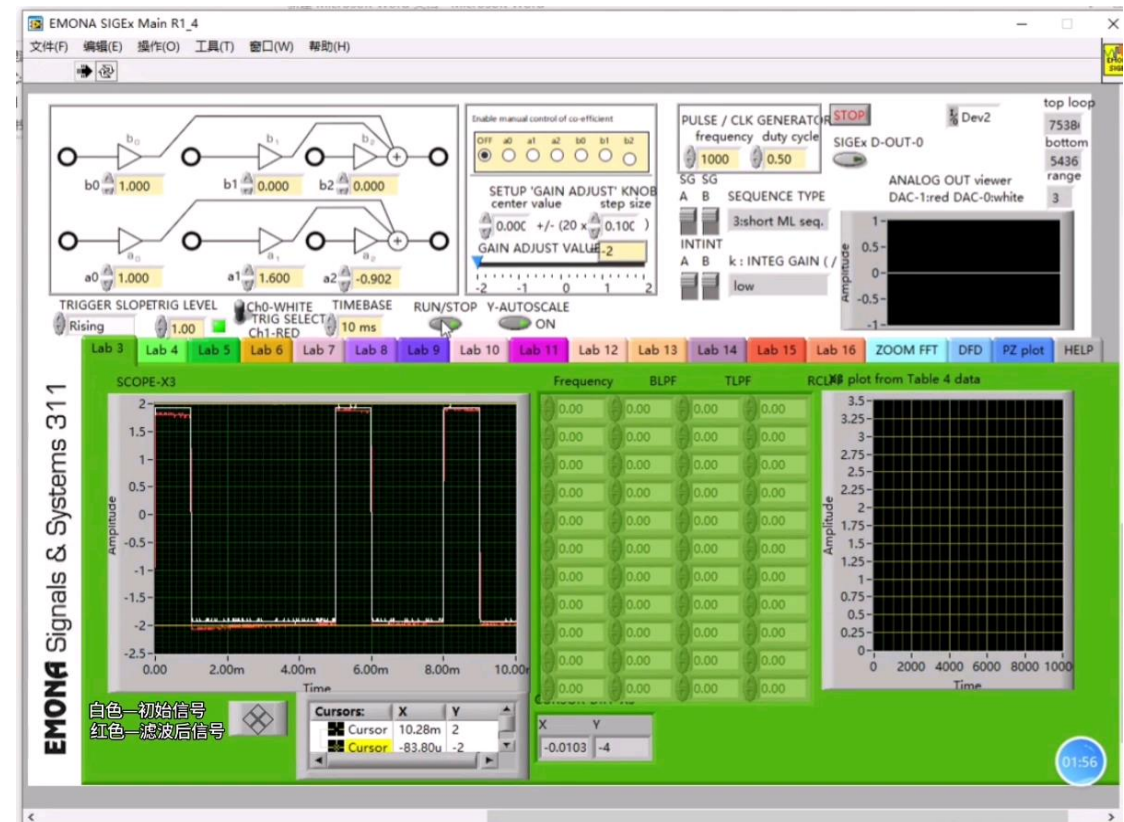
5.1 Introduction



- Rectangular wave passing through RC low-pass filters with different bandwidths



Seriously distorted waveform



Little distorted waveform

5.2 Frequency Response of Systems



Definition and Physical Meaning of Frequency-Domain System Function (Frequency Response)

Definition: $H(\omega) = \frac{R(\omega)}{E(\omega)}$ or $H(\omega) = F[h(t)]$

Fourier Transform of the Response:

$$R(\omega) = H(\omega)E(\omega) = |H(\omega)|e^{j\varphi(\omega)}E(\omega)$$

Roles of Frequency Response: Changing the Spectrum of the Input Signal

1. Weight each frequency component of the signal; some frequency components are enhanced, while others are relatively weakened or unchanged.
2. Each frequency component generates its own phase shift during transmission.

Frequency response can intuitively reflect the change of the signal spectrum after passing through the system, and explain the difference between excitation and response time-domain waveforms, with **clear physical concepts**.

5.2 Frequency Response of Systems



When a **complex exponential** signal is the excitation of an LTI system, the **steady-state response is a complex exponential function of the same frequency**, with **only the amplitude and phase changed**.

Proof:

$$y(t) = h(t) * e^{j\omega_0 t} = \int_{-\infty}^{\infty} h(\tau) e^{j\omega_0(t-\tau)} d\tau = e^{j\omega_0 t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega_0 \tau} d\tau = H(\omega_0) e^{j\omega_0 t} = |H(\omega_0)| e^{j[\omega_0 t + \varphi(\omega_0)]}$$

Block diagram for complex exponential excitation:

$$e^{j\omega_0 t} \rightarrow \boxed{h(t)} \rightarrow y(t) = e^{j\omega_0 t} * h(t) = |H(\omega_0)| e^{j[\omega_0 t + \varphi(\omega_0)]}$$

Block diagram for complex exponential excitation (negative frequency):

$$e^{-j\omega_0 t} \rightarrow \boxed{h(t)} \rightarrow y(t) = e^{-j\omega_0 t} * h(t) = |H(\omega_0)| e^{-j[\omega_0 t + \varphi(\omega_0)]}$$

When a **sinusoidal** signal is the excitation of an LTI system, **the steady-state response is a sinusoidal function of the same frequency**, with **only the amplitude and phase changed**.

Block diagram for sinusoidal excitation (cosine):

$$\cos(\omega_0 t) \rightarrow \boxed{h(t)} \rightarrow y(t) = \cos(\omega_0 t) * h(t) = |H(\omega_0)| \cos[\omega_0 t + \varphi(\omega_0)]$$

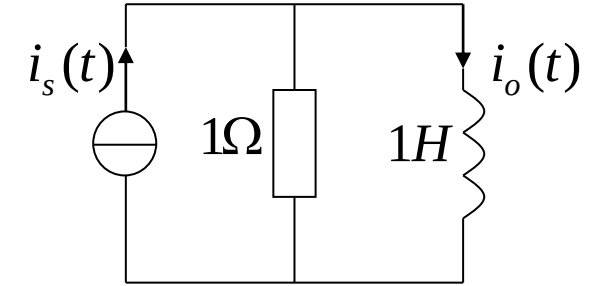
Block diagram for sinusoidal excitation (sine):

$$\sin(\omega_0 t) \rightarrow \boxed{h(t)} \rightarrow y(t) = \sin(\omega_0 t) * h(t) = |H(\omega_0)| \sin[\omega_0 t + \varphi(\omega_0)]$$

5.2 Frequency Response of Systems



Example 5-1: An RL circuit is shown in the figure. The excitation is a current source $i_s(t)$, and the response is the current $i_o(t)$ in the inductor. List the input-output differential equation of the circuit, find its frequency response. If the excitation $i_s(t) = \cos t$, find the sinusoidal steady-state response.



Solution: The system equation is: $i_R(t) + i_o(t) = i_s(t)$

$$\therefore \frac{L}{R} \frac{di_o(t)}{dt} + i_o(t) = i_s(t) \qquad \frac{di_o(t)}{dt} + i_o(t) = i_s(t)$$

Take the Fourier Transform on both sides:

$$(j\omega + 1)I_o(\omega) = I_s(\omega)$$

Thus, $H(\omega) = \frac{I_o(\omega)}{I_s(\omega)} = \frac{1}{j\omega + 1}$ $H(1) = H(\omega)|_{\omega=1} = \frac{1}{j+1} = \frac{1}{\sqrt{2}} e^{-j\frac{\pi}{4}}$

$$i_o(t) = \frac{1}{\sqrt{2}} \cos\left(t - \frac{\pi}{4}\right)$$

To make $H(s)|_{s=j\omega} = H(\omega)$, the poles of the system function $H(s)$ should be located on the () of the s-plane.

- ☒ A Left half-plane
- ☐ B Right half-plane
- ☐ C Imaginary axis

提交

5.2 Frequency Response of Systems



Using System Function to Find Frequency Response

(1) $H(\omega)$ describes the frequency response characteristic of sinusoidal steady-state response.

$$H(s) \big|_{s=j\omega} = H(\omega) = |H(\omega)| e^{j\varphi(\omega)}$$

(2) $H(\omega)$ is the system function, the Fourier transform of the system impulse response $h(t)$.

$$H(\omega) = F[h(t)]$$

Condition for their equality: All the poles of $H(s)$ are located on the left half-plane of the s-plane (stable system).

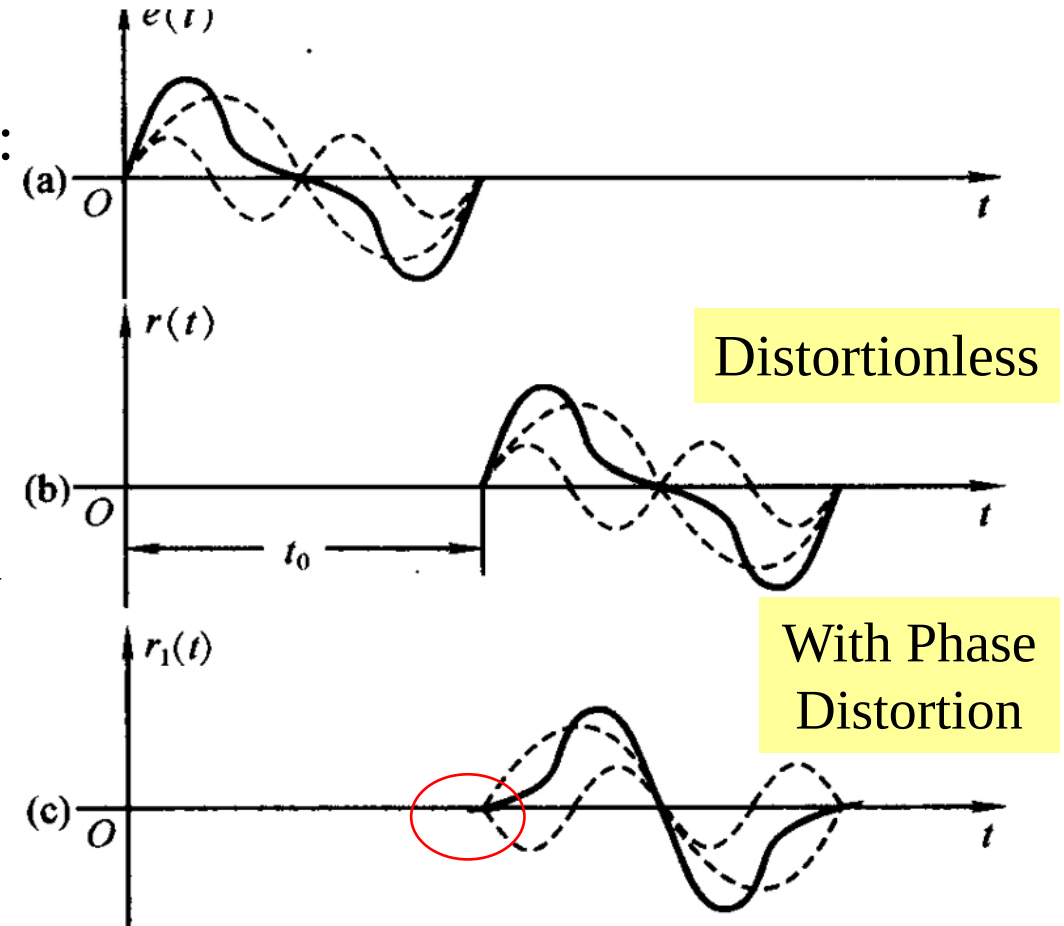
$$F[h(t)] = H(\omega) = H(s) \big|_{s=j\omega}$$

5.3 Distortionless Transmission

5.3.1 Causes of Signal Distortion

Observed in the frequency domain, signal distortion caused by linear systems has two factors:

- (1) **Amplitude Distortion**: The system **attenuates the amplitude of each frequency component in the signal to different degrees**, changing the relative amplitude of the response components.
- (2) **Phase Distortion**: The phase shift generated by the system for each frequency component **is not proportional to the frequency**, changing the relative position of each frequency component of the response on the time axis.



Neither of them generates new frequency components, so there is **no frequency distortion**.

5.3 Distortionless Transmission

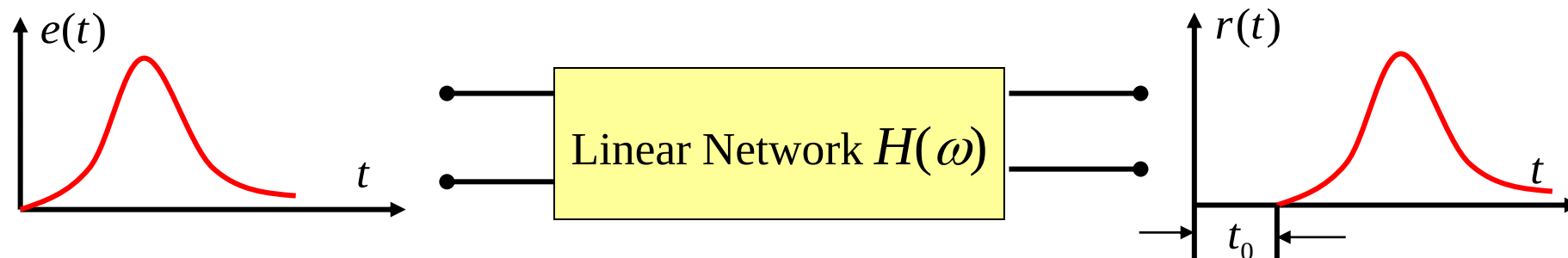
5.3.2 Concept of Distortionless Signal Transmission

Distortionless transmission: Compared with the excitation signal, the response signal only differs in **magnitude and occurrence time** without any change in waveform (**the time-domain waveform remains unchanged during transmission**), i.e.,

$$r(t) = Ke(t - t_0)$$

K is a constant and t_0 is the time delay.

When the distortionless condition is satisfied, the waveform of $r(t)$ is a delay of the waveform of $e(t)$ by t_0 time.



5.3 Distortionless Transmission

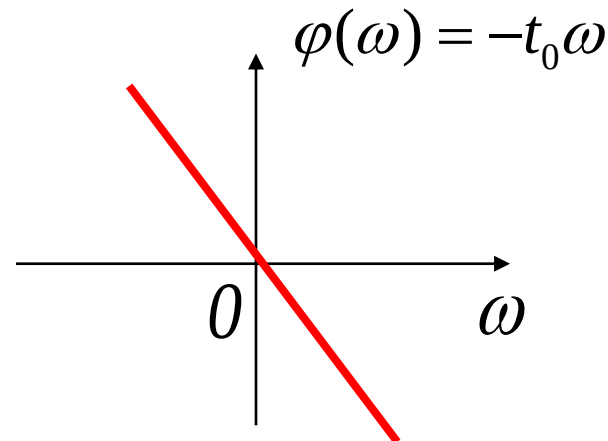
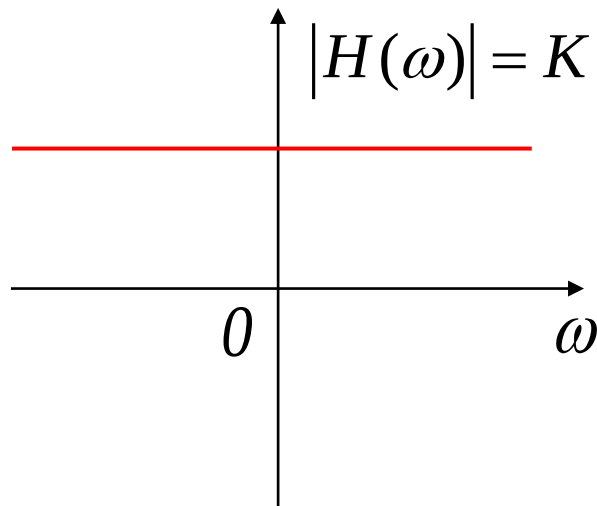
5.3.3 Conditions for Distortionless Signal Transmission in Systems

1. Frequency-Domain Condition for Distortionless Transmission of Systems

$$r(t) = Ke(t - t_0)$$

Take the Fourier transform on both sides: $R(\omega) = KE(\omega)e^{-j\omega t_0}$

$$H(\omega) = \frac{R(\omega)}{E(\omega)} = Ke^{-j\omega t_0} \quad \text{where} \quad |H(\omega)| = K, \quad \varphi(\omega) = -\omega t_0$$



The **magnitude-frequency response** of the system is a constant **K**;

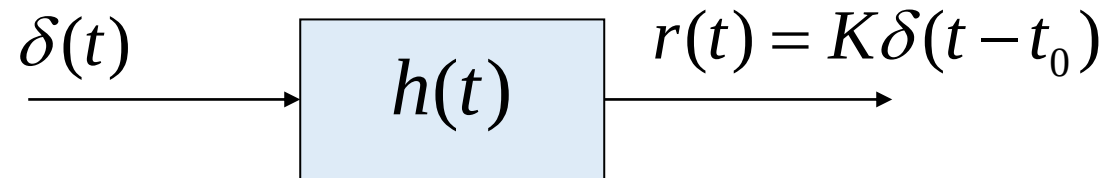
The **phase-frequency response** is a straight line passing through the origin with a slope of **$-t_0$**

2. Time-Domain Condition for Distortionless Transmission of Systems

$$H(\omega) = Ke^{-j\omega t_0}$$

Take the inverse Fourier transform: $h(t) = K\delta(t - t_0)$

That is, the system's impulse response is also an impulse function, delayed by t_0 .



Impulse signals are commonly used to test the fidelity of systems.

A distortionless transmission system is equivalent to a ().

- ☐ A Low-pass network
- ☐ B High-pass network
- ☐ C Band-pass network
- ☒ D All-pass network

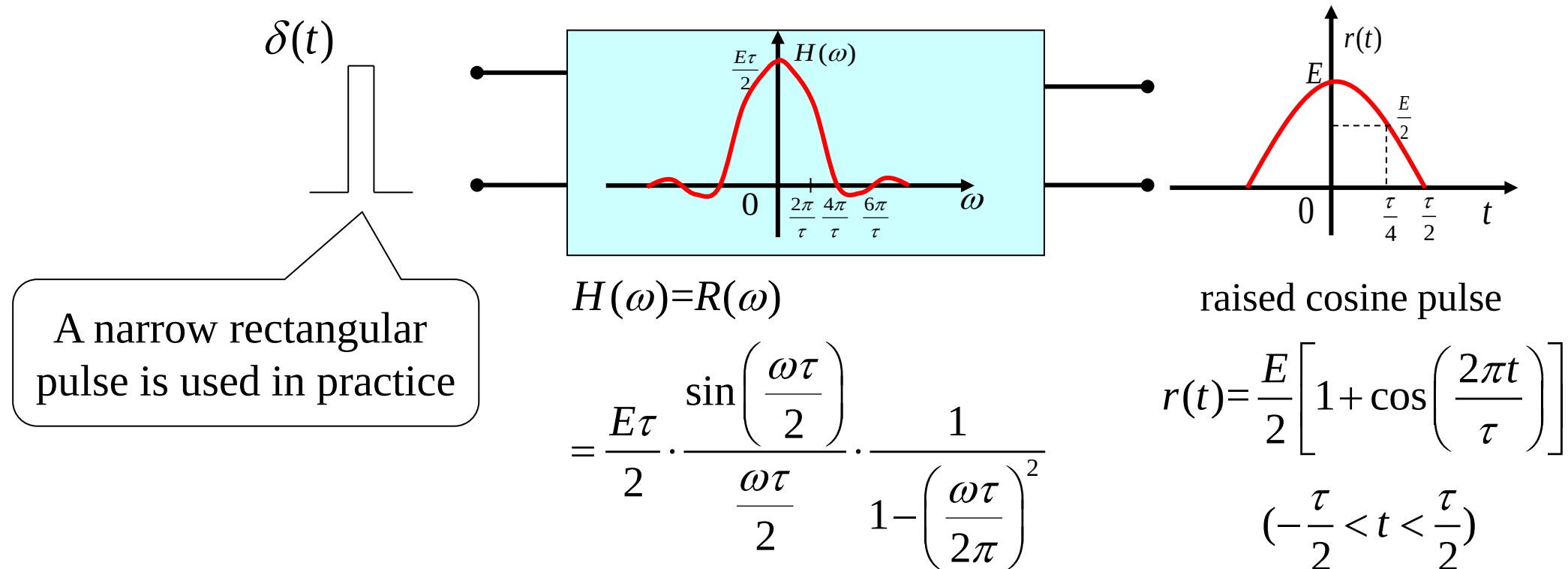
提交

5.3 Distortionless Transmission

5.3.4 Generating Specific Waveforms Using Distortion

In practice, **specific waveforms can be formed by designing system functions and utilizing signal distortion.**

To generate a specific waveform $r(t)$ by applying an impulse signal to the system:



5.4 Ideal Low-Pass Filter

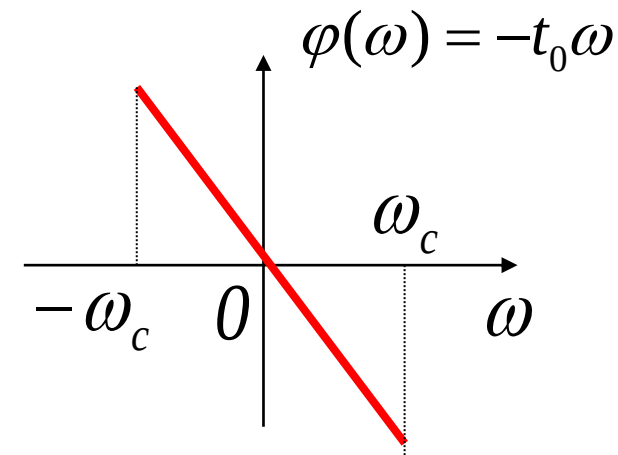
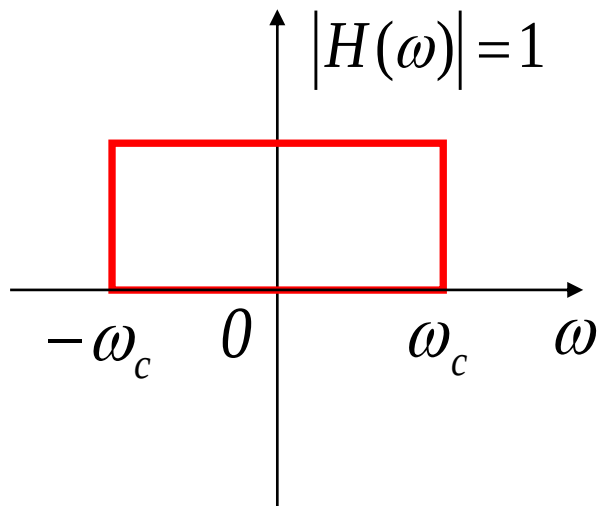
5.4.1 Frequency-Domain Characteristics and Impulse Response of the Ideal LPF

Ideal Filter: Its frequency response has a **rectangular amplitude characteristic** and a **linear phase shift characteristic**.

Frequency response of an ideal low-pass filter (LPF):

$$H(\omega) = |H(\omega)| e^{j\varphi(\omega)} = \begin{cases} 1 \cdot e^{-j\omega t_0} & |\omega| < \omega_c \\ 0 & \text{other} \end{cases}$$

ω_c is called the **cutoff frequency**.



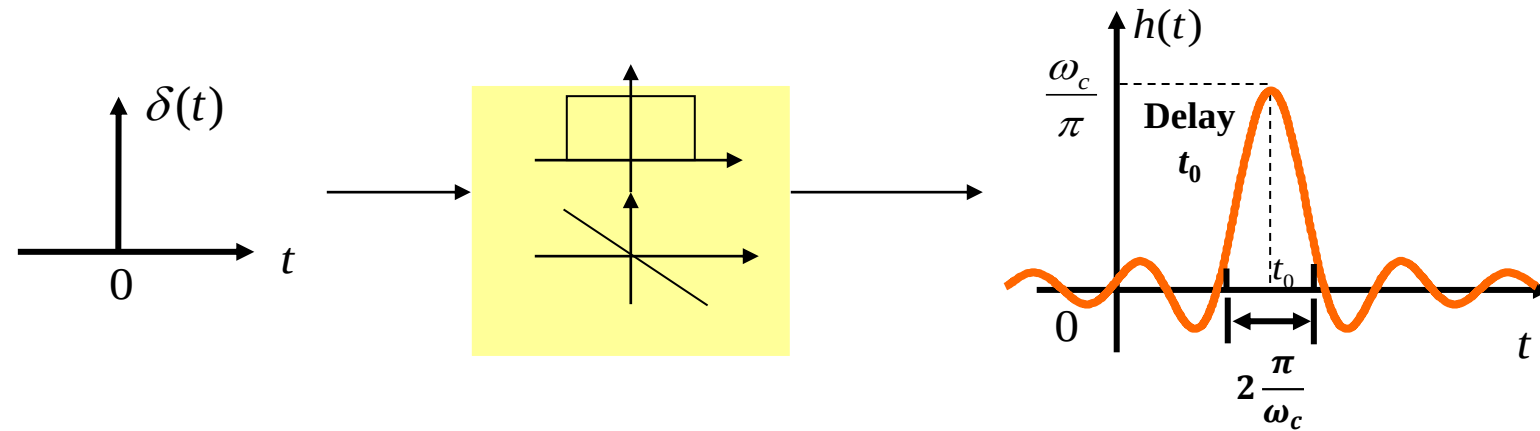
Find all the FALSE statements about the ideal LPF ():

- ☐ A The impulse function is distorted when passing through an ideal low-pass filter.
- ☒ B The impulse function passes through an ideal low-pass filter without distortion.
- ☒ C The ideal low-pass filter is physically realizable.
- ☐ D The ideal low-pass filter is physically unrealizable.

提交

5.4 Ideal Low-Pass Filter

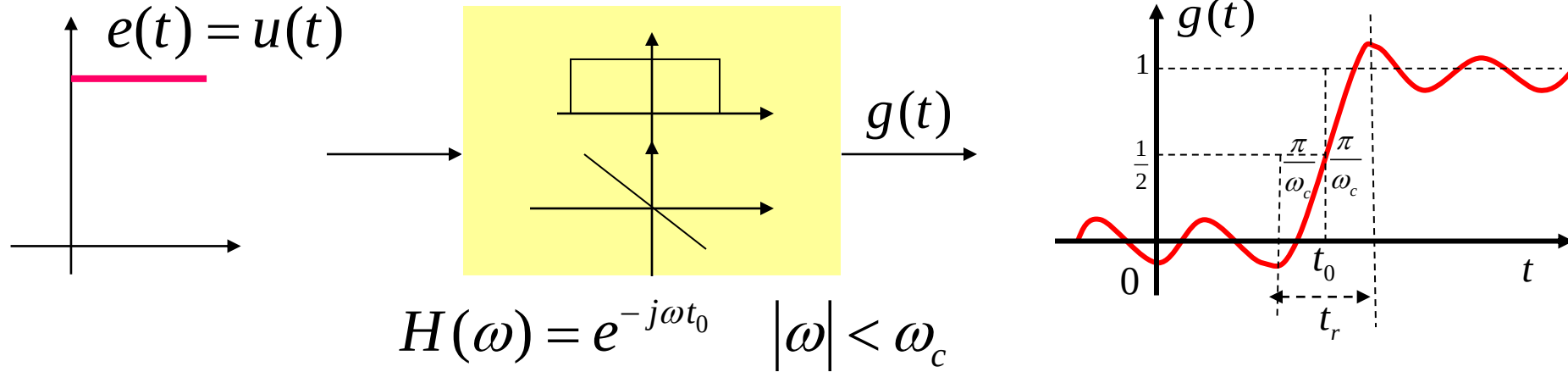
Impulse Response of the Ideal Low-Pass Filter



- (1) $h(t)$ is a **sampling function** (i.e., **Sa function**), which differs from the waveform of the input signal $\delta(t)$, so **distortion occurs**.
- ✓ Reason: The ideal low-pass filter is a band-limited system, while the frequency bandwidth of the impulse signal $\delta(t)$ is infinite.
- (2) The peak of $h(t)$ is delayed by a time t_0 . t_0 is the **group delay** of the **ideal low-pass**.
- (3) $h(t)$ has output when $t < 0$, indicating that the **ideal low-pass filter** is a **non-causal system** and thus **physically unrealizable**.

5.4 Ideal Low-Pass Filter

5.4.2 Step Response of Ideal Low-Pass Filter



$$E(\omega) = F[u(t)] = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$G(\omega) = H(\omega)E(\omega) = \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] e^{-j\omega t_0} \quad -\omega_c < \omega < \omega_c$$

$$g(t) = F^{-1}[G(\omega)] = \frac{1}{2} + \frac{1}{\pi} \int_0^{\omega_c} \frac{\sin[\omega(t-t_0)]}{\omega} d\omega = \frac{1}{2} + \frac{1}{\pi} Si[\omega_c(t-t_0)]$$

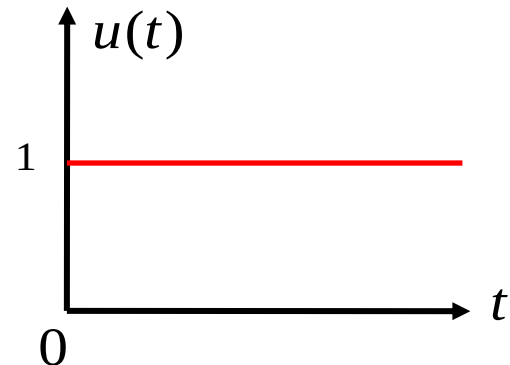
where $Si(y) = \int_0^y \frac{\sin x}{x} dx = \int_0^y Sa(x) dx$ Letting $x = \omega(t-t_0)$

5.4 Ideal Low-Pass Filter

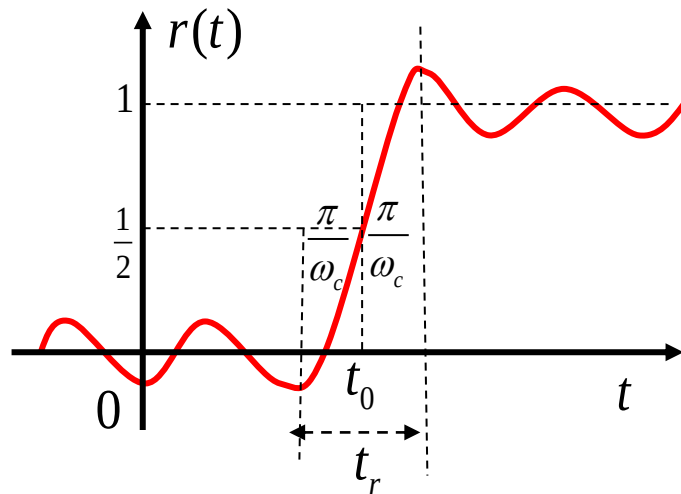


Let $x = \omega(t - t_0)$

then
$$r(t) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\omega_c(t-t_0)} \frac{\sin x}{x} dx = \frac{1}{2} + \frac{1}{\pi} \text{Si}[\omega_c(t - t_0)]$$



Waveform of Input Signal



Analysis:

- (1) $g(t)$ is delayed by time t_0 compared to $u(t)$.
- (2) Rising time: The time required for the output to go from its minimum to maximum value.

$$t_r = 2 \cdot \frac{\pi}{\omega_c}$$

The rise time of $g(t)$ corresponds to the pulse width of $h(t)$. With $t_r \omega_c = 2\pi$, the rising time is inversely proportional to bandwidth. This embodies the **Uncertainty Principle**--a system cannot achieve optimal time and frequency resolution at the same time.

- (3) The Gibbs phenomenon exists, and the overshoot amplitude in the waveform is close to 9%.

5.4 Ideal Low-Pass Filter

5.4.3 Rectangular Pulse Response of Ideal Low-Pass Filter

$$e(t) = u(t) - u(t - \tau)$$

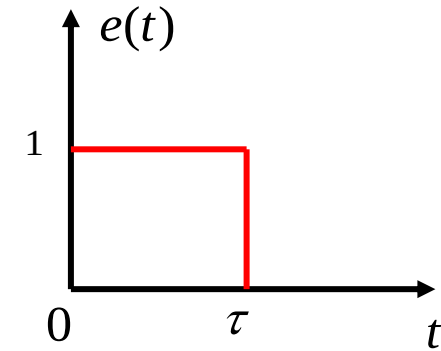
$$r(t) = \left\{ \frac{1}{2} + \frac{1}{\pi} \text{Si}[\omega_c(t - t_0)] \right\} - \left\{ \frac{1}{2} + \frac{1}{\pi} \text{Si}[\omega_c(t - \tau - t_0)] \right\}$$

$$r(t) = \frac{1}{\pi} \{ \text{Si}[\omega_c(t - t_0)] - \text{Si}[\omega_c(t - t_0 - \tau)] \}$$

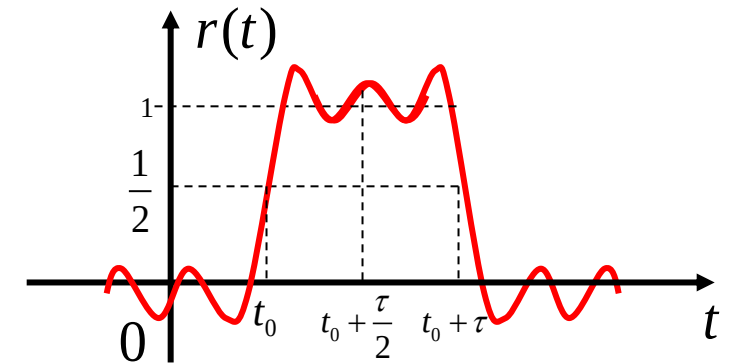
The bandwidth of the input signal is $B = 2\pi/\tau$.

If the pulse width $\tau \gg \frac{2\pi}{\omega_c}$ ($\omega_c \gg B$), the response waveform is roughly rectangular;

If the pulse width $\tau \leq \frac{2\pi}{\omega_c}$, the response waveform is severely distorted, resembling a sine wave.

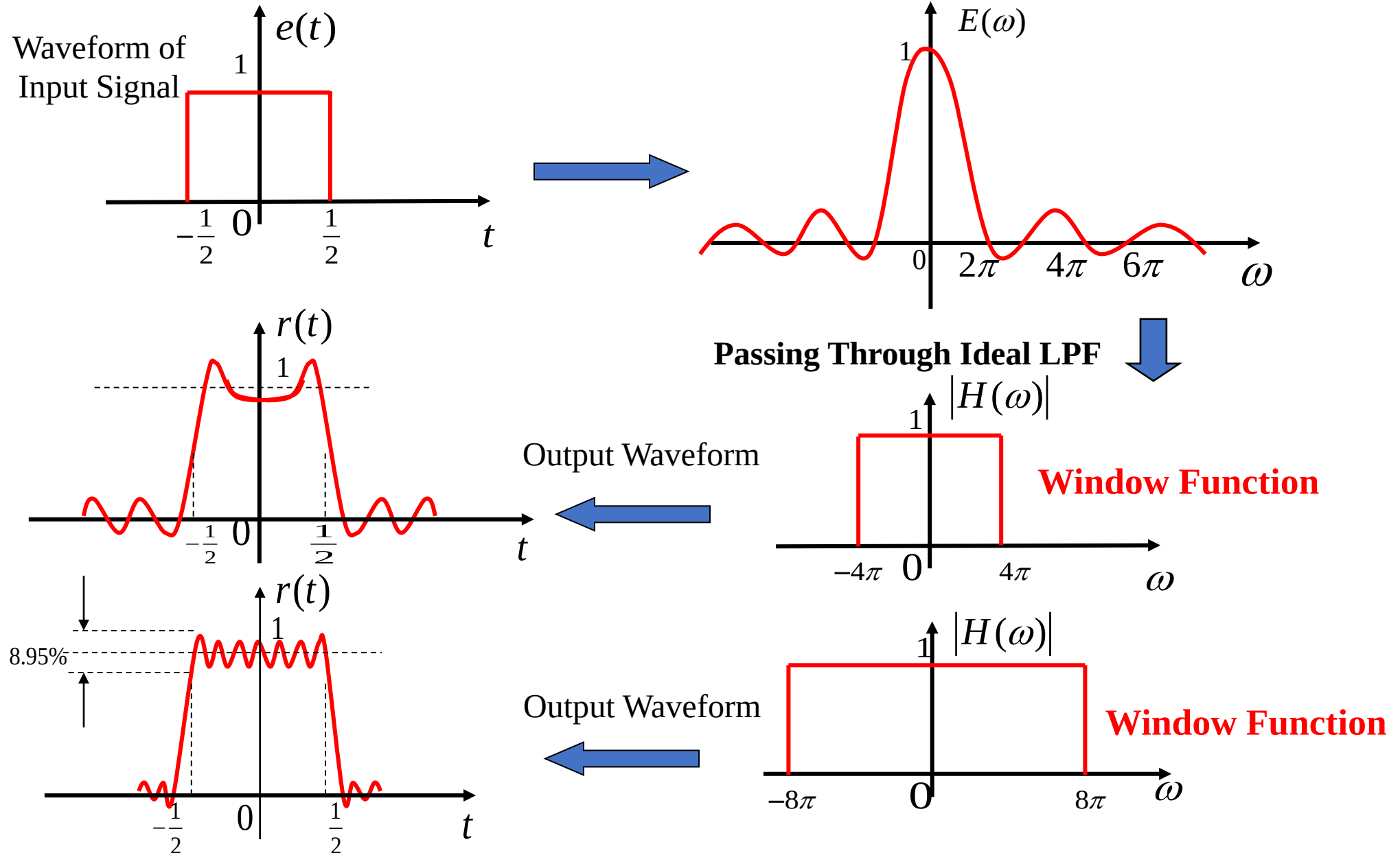


Waveform of Input Signal



Output Signal Waveform of Ideal Low-Pass Filter

5.4 Ideal Low-Pass Filter



5.5 Physical Realizability of Systems

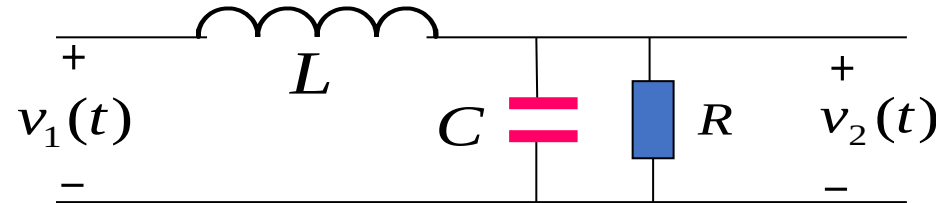


Ideal low-pass, band-pass, high-pass, and band-stop filters are all **physically unrealizable**.
What kind of system mathematical model can be physically realized?

Question: Roughly sketch the amplitude-frequency characteristics of the **ideal low-pass filter** and the **physically realizable low-pass filter** respectively.

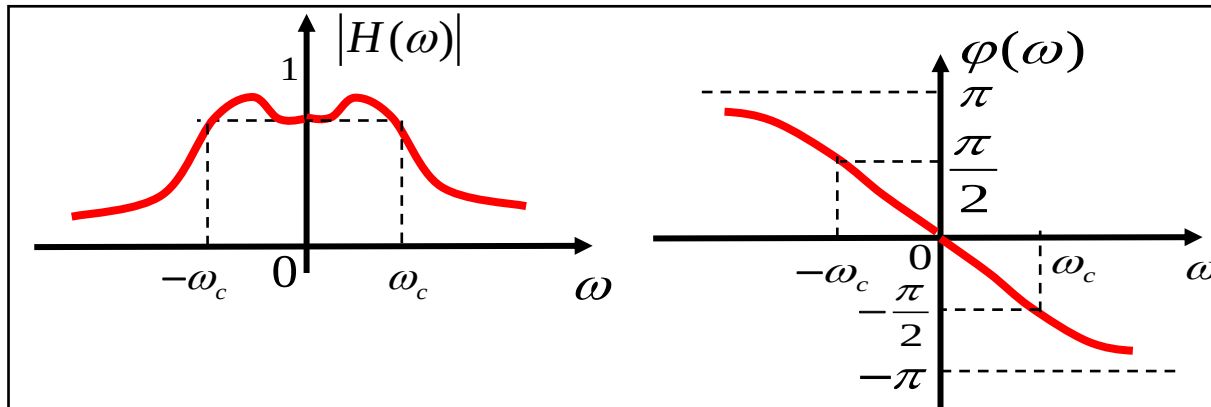
5.5 Physical Realizability of Systems

Example 5-2: A low-pass filter circuit is shown. Given $R = \sqrt{L/C}$, find $h(t)$.

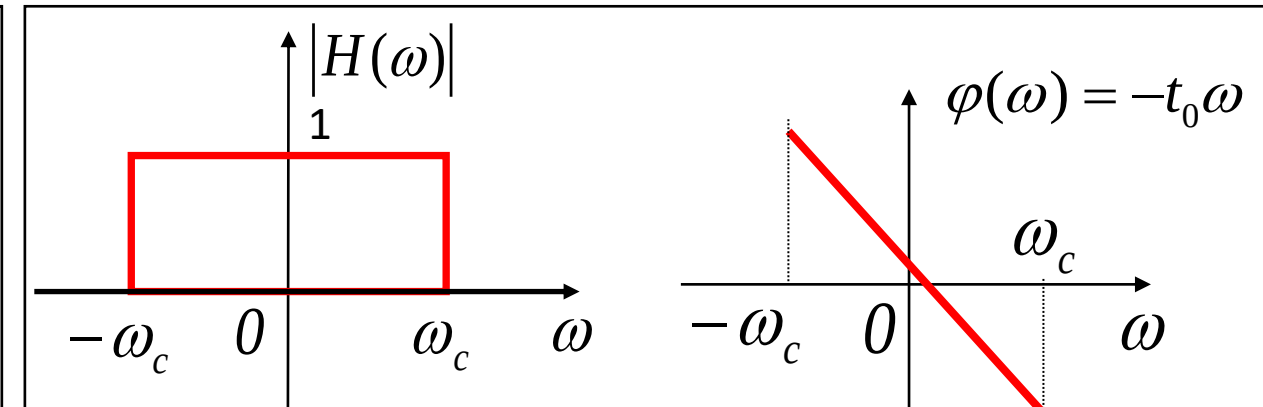


Solution:
$$H(\omega) = \frac{1}{1 - \left(\frac{\omega}{\omega_c}\right)^2 + j\frac{\omega}{\omega_c}} = \frac{2\omega_c}{\sqrt{3}} \frac{\frac{\sqrt{3}}{2}\omega_c}{\left(\frac{\omega_c}{2} + j\omega\right)^2 + \left(\frac{\sqrt{3}}{2}\omega_c\right)^2}$$

$$\omega_c = \frac{1}{\sqrt{LC}}$$



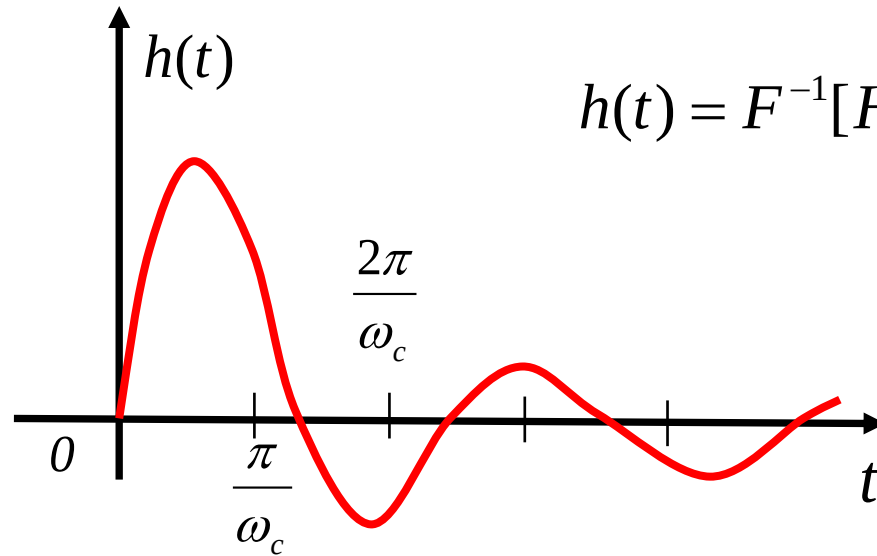
Frequency Response of Physically Realizable LPF



Ideal LPF's Frequency Response

5.5 Physical Realizability of Systems

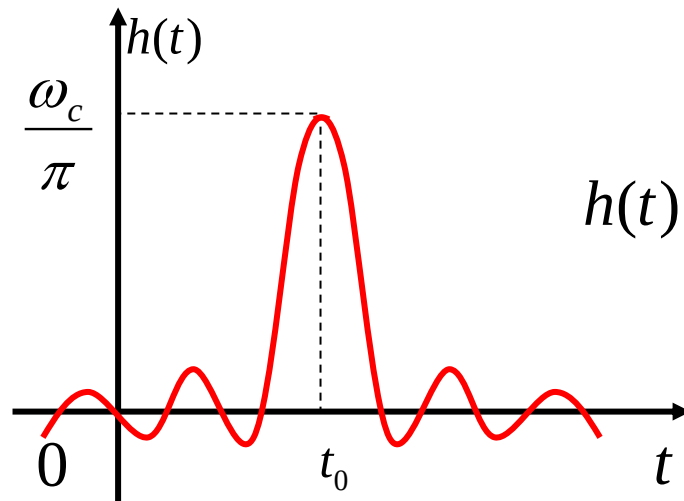
Impulse Response of Physically Realizable Filter



$$h(t) = F^{-1}[H(\omega)] = \frac{2\omega_c}{\sqrt{3}} e^{-\frac{\omega_c}{2}t} \sin\left(\frac{\sqrt{3}}{2}\omega_c t\right) \quad (t > 0)$$

It is composed of actual circuits and starts at $t > 0$, so it is physically realizable.

Impulse Response of Ideal Low-Pass Filter



$$h(t) = \frac{\omega_c}{\pi} \text{Sa}[\omega_c(t - t_0)]$$

5.5 Physical Realizability of Systems

Criteria for Realizable and Unrealizable Systems

(1) **Time-Domain**——**Causality** (the output only relates to the present and past of the input) is a **necessary and sufficient condition** for physical realizability, $h(t) = 0 \ (t < 0)$

(2) **Frequency-Domain**——If the squared magnitude frequency response is integrable:

$$\int_{-\infty}^{\infty} |H(\omega)|^2 d\omega < \infty$$

The Paley-Wiener Criterion —— a **necessary but not sufficient condition** for physical realizability (causality)

$$\int_{-\infty}^{\infty} \frac{|\ln |H(\omega)||}{1 + \omega^2} d\omega < \infty$$

It does not impose constraints on phase characteristics.

If this condition is not satisfied, the system is unrealizable.

$|H(\omega)|$ is only allowed to be zero at some discontinuous frequency points; it is not allowed to be zero over a finite frequency band. That is, the attenuation speed of the amplitude characteristic is limited. Ideal low-pass, ideal high-pass, ideal band-pass, and ideal band-stop filters are all unrealizable.

$$\text{If } |H(\omega)| = 0, \quad |\ln |H(\omega)|| \rightarrow \infty$$

5.5 Physical Realizability of Systems



The Paley-Wiener criterion is a necessary but not sufficient condition for physical realizability.

$h(t) \rightarrow H(\omega)$ Causal system

$h(t + t_0) \rightarrow H(\omega)e^{j\omega t_0}, \quad t_0 > 0$

Non – causal system

Both have the same amplitude response (satisfying the criterion) but different phase response.

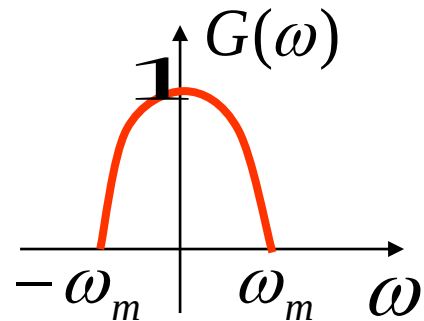
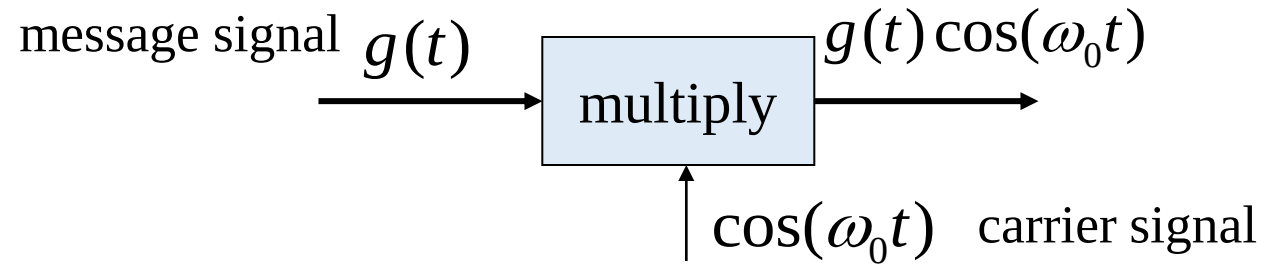
If $|H(\omega)|$ satisfies the criterion, an appropriate phase function $\varphi(\omega)$ can be paired with it to form a physically realizable system function.

5.6.1 Purposes of Modulation

- **Condition for Effective Signal Radiation:** The antenna size should be one-tenth or larger of the wavelength of the radiated signal. For example, for a voice signal, the antenna size would need to be dozens of kilometers, which is practically impossible to manufacture.
- **Modulation:** Shifts the signal spectrum to any desired higher frequency range, enabling the production of antennas with reasonable sizes for easy transmission.
- **Essence of Modulation:** Shifts the spectra of various signals so that they occupy different frequency ranges without overlapping. That is, signals are attached to carriers of different frequencies, allowing the receiver to separate the desired frequency signal without mutual interference (Frequency-Shifting Theorem).
- **More Attenuation in Radio Wave Propagation:** High-frequency signals are prone to attenuation.
- **Multiplexing:** Use the same radio station to shift the spectra of different signals to different frequency bands, enabling multi-channel communication within the same channel.

5.6 Modulation and Demodulation

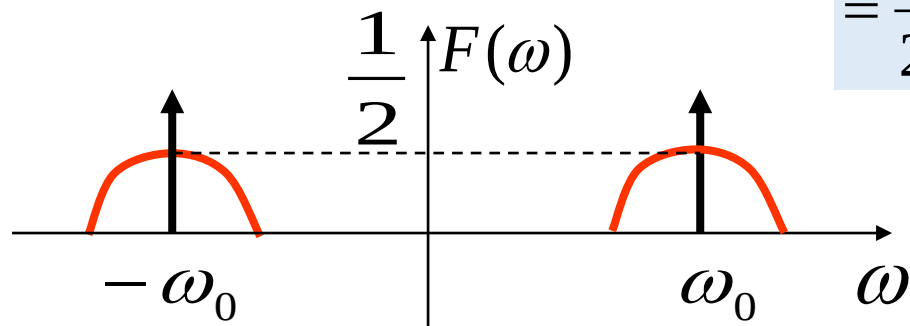
5.6.2 Modulation Principle



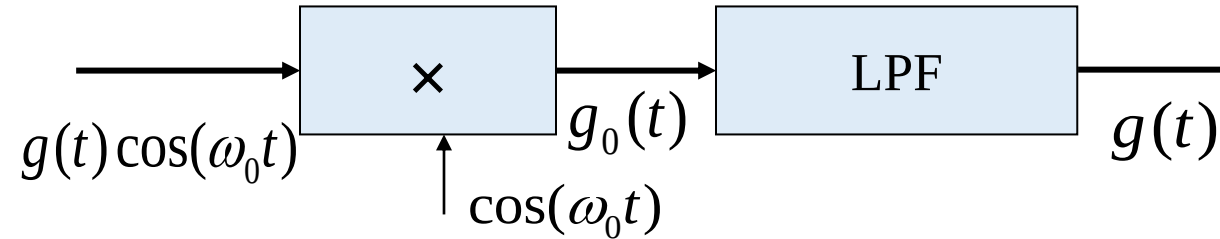
modulated signal $f(t) = g(t) \cos \omega_0 t$

$$F(\omega) = \frac{1}{2\pi} G(\omega) * [\pi\delta(\omega + \omega_0) + \pi\delta(\omega - \omega_0)]$$

$$= \frac{1}{2} [G(\omega + \omega_0) + G(\omega - \omega_0)]$$



5.6.3 Synchronous Demodulation Principle



Synchronous Demodulation: The receiver has a local carrier that is frequency- and phase-synchronized with the transmitter's carrier.

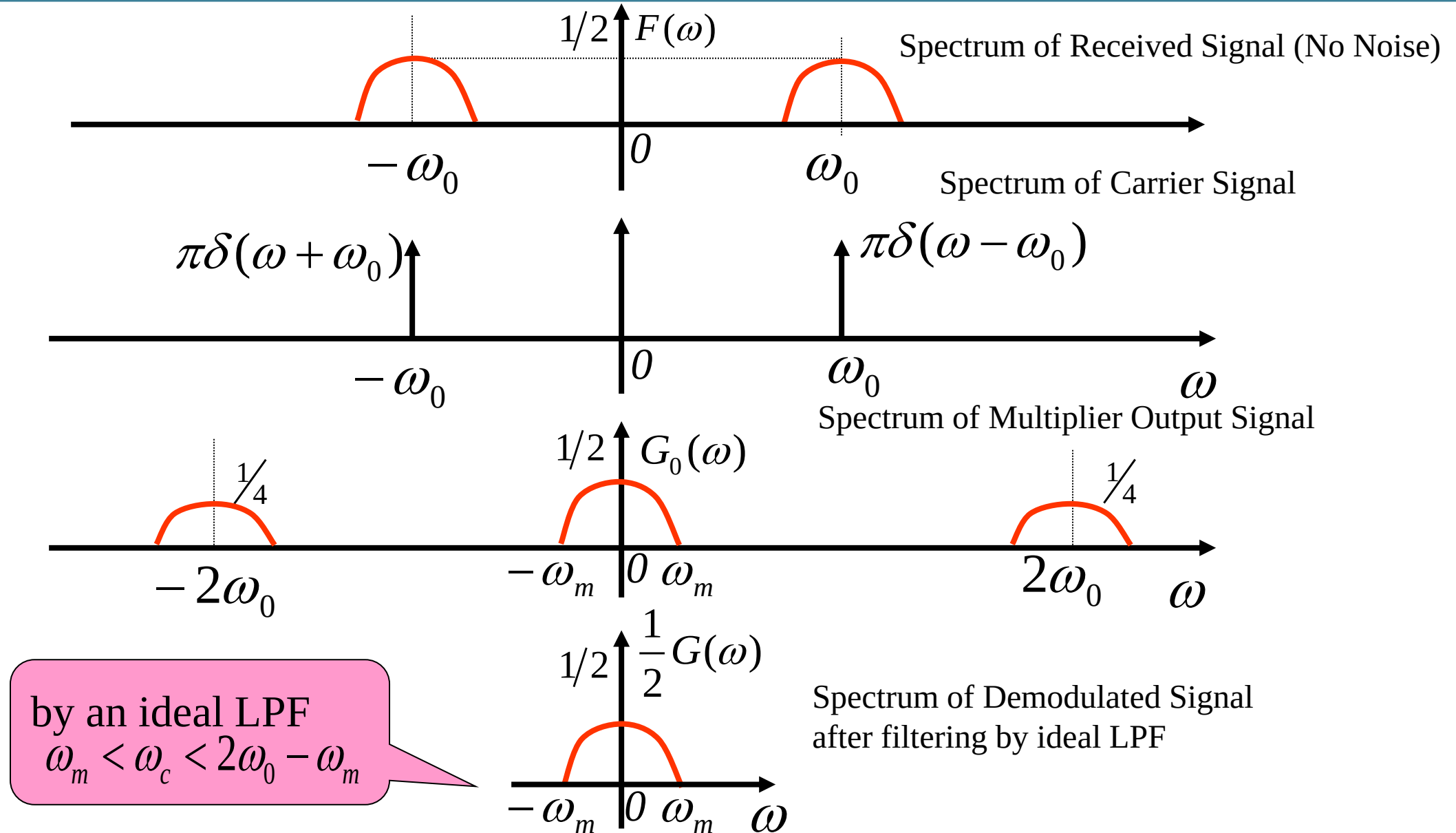
$$g_0(t) = [g(t)\cos\omega_0 t]\cos\omega_0 t$$

$$= \frac{1}{2}g(t)(1 + \cos 2\omega_0 t)$$

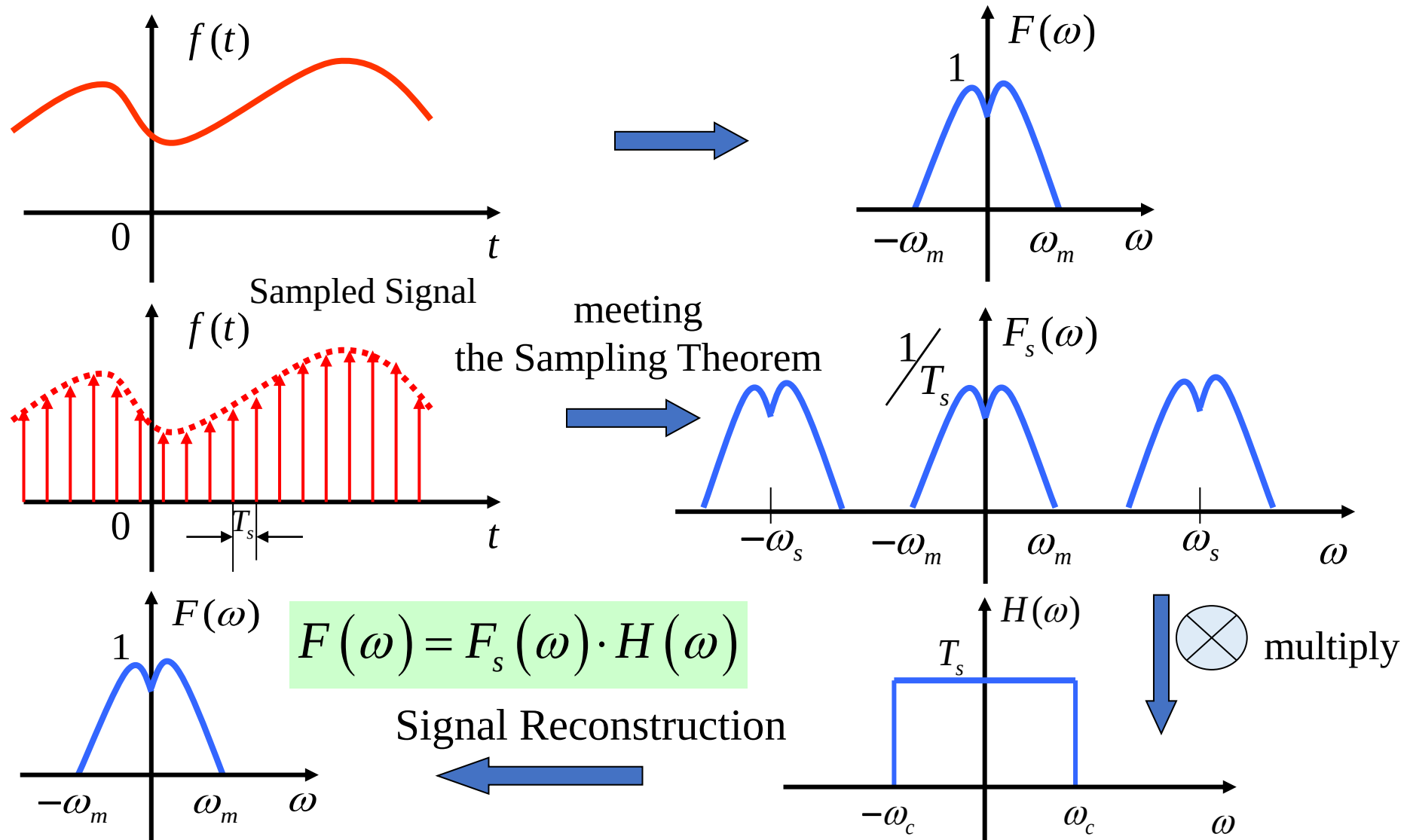
$$= \frac{1}{2}g(t) + \frac{1}{2}g(t)\cos 2\omega_0 t$$

$$G_0(\omega) = \frac{1}{2}G(\omega) + \frac{1}{4}[G(\omega + 2\omega_0) + G(\omega - 2\omega_0)]$$

5.6 Modulation and Demodulation

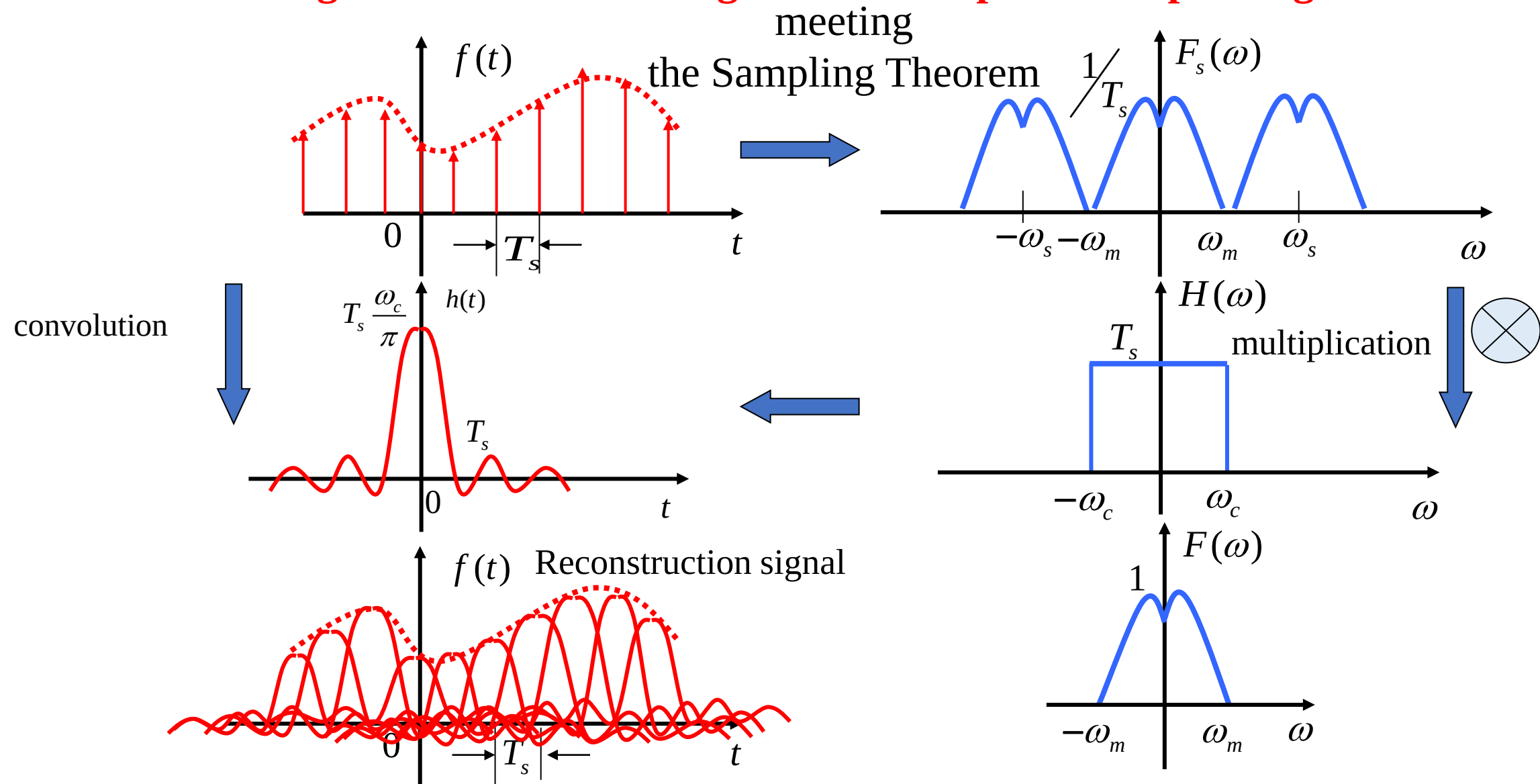


5.7 Recovering Continuous-Time Signals from Sampled Signals



5.7 Recovering Continuous-Time Signals from Sampled Signals

Recovering Continuous-Time Signals from Impulse-Sampled Signals



5.7 Recovering Continuous-Time Signals from Sampled Signals



Time-Domain Analysis

$$f_s(t) = f(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} f(nT_s) \delta(t - nT_s)$$

$$F_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s) \quad \omega_s = \frac{2\pi}{T_s}$$

$$\therefore f(t) = h(t) * f_s(t) = T_s \cdot \frac{\omega_c}{\pi} \text{Sa}(\omega_c t) * \sum_{n=-\infty}^{\infty} f(nT_s) \delta[\omega_c(t - nT_s)]$$

$$= T_s \cdot \frac{\omega_c}{\pi} \sum_{n=-\infty}^{\infty} f(nT_s) \text{Sa}[\omega_c(t - nT_s)]$$

Interpolation Formula – Interpolate between Sampling Points to Recover Continuous-Time Signals.

The continuous signal $f(t)$ is expanded into an infinite series of sampling functions, with coefficients proportional to $f(nT_s)$. (linear superposition)

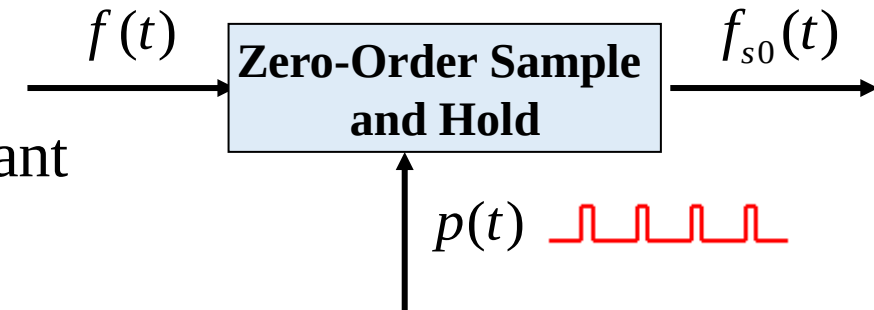
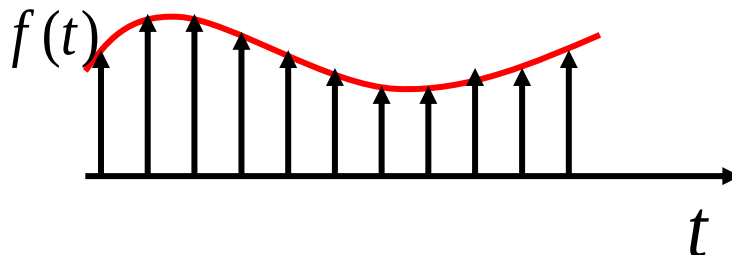
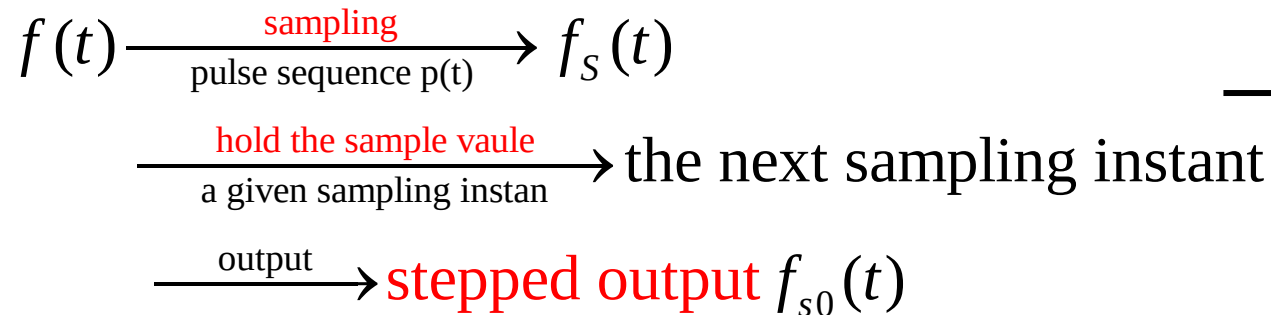
Critical Sampling, $\omega_s = 2\omega_m, T_s = \pi / \omega_m, \omega_c = \omega_m$

$$\therefore f(t) = \sum_{n=-\infty}^{\infty} f(nT_s) \text{Sa}[\omega_c(t - nT_s)]$$

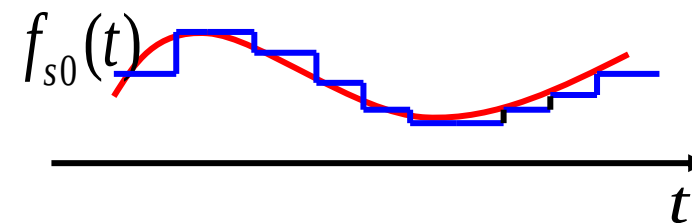
5.7 Recovering Continuous-Time Signals from Sampled Signals

Sample and Hold

Since generating and transmitting ideal sampling pulses is challenging in practical circuits, alternative sampling methods are employed in communication systems. The most common one is **zero-order sample and hold**, simply referred to as **sample and hold**.



Block Diagram of Zero-Order Sample and Hold



5.7 Recovering Continuous-Time Signals from Sampled Signals



Sample and Hold: Construct a linear system to achieve the level-holding function

$$H_0(\omega) = T_s \text{Sa}\left(\frac{\omega T_s}{2}\right) e^{-j\frac{\omega T_s}{2}}$$

$$F_{s0}(\omega) = F_s(\omega) H_0(\omega) = \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s) \text{Sa}\left(\frac{\omega T_s}{2}\right) e^{-j\frac{\omega T_s}{2}}$$

To recover the signal $f(t)$, **a low-pass filter $H_{0r}(\omega)$ with compensatory properties is used.**

$$\xrightarrow{\text{recover}} F(\omega) = F_{s0}(\omega) \cdot H_{0r}(\omega) \xrightarrow{\mathcal{F}^{-1}} f(t)$$

$$H_{0r}(\omega) = \begin{cases} \frac{e^{j\frac{\omega T_s}{2}}}{\text{Sa}\left(\frac{\omega T_s}{2}\right)} & |\omega| \leq \frac{\omega_s}{2} \\ 0 & |\omega| \geq \frac{\omega_s}{2} \end{cases}$$

Generally, it is only required that the $|H_{0r}(\omega)|$ curve is approximately compensatory, and $\varphi(\omega)$ has a linear phase – shift characteristic.

5.8.1 Multiplexing

Combining several signals in a certain manner and transmitting them collectively over the same channel is defined as multiplexing.

5.8.2 Frequency Division Multiplexing (FDM)

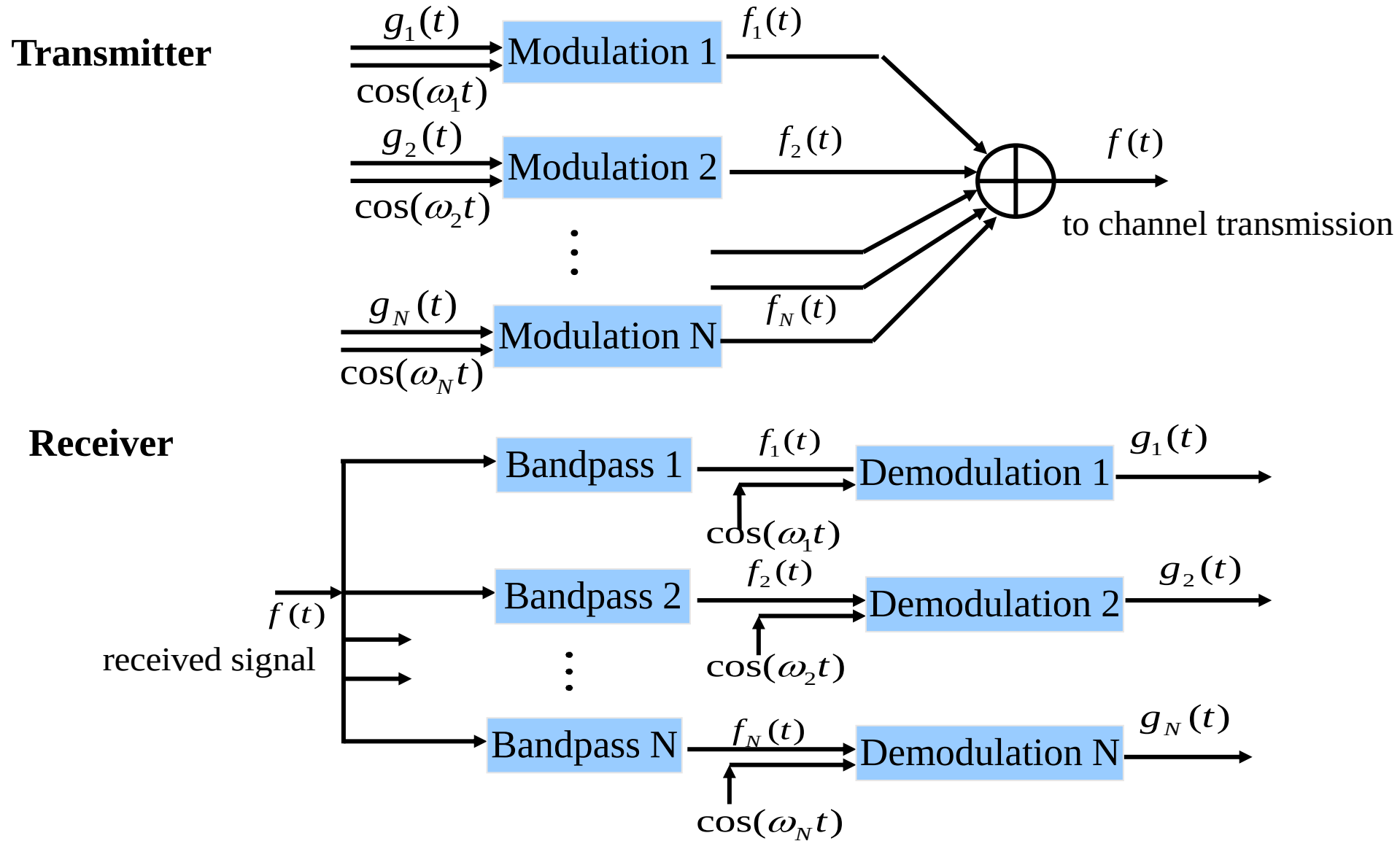
It is based on the frequency-shifting property of Fourier transform:

- (1) At the transmitting end, the spectra of each signal are shifted to distinct frequency ranges, making them **non-overlapping in the spectrum**. This allows multiplexed transmission over a single channel.
- (2) At the receiving end, multiple filters are used to separate each signal. After demodulation, the original signals of each channel can be restored.

5.8 Frequency-Division Multiplexing (FDM) & Time-Division Multiplexing (TDM)

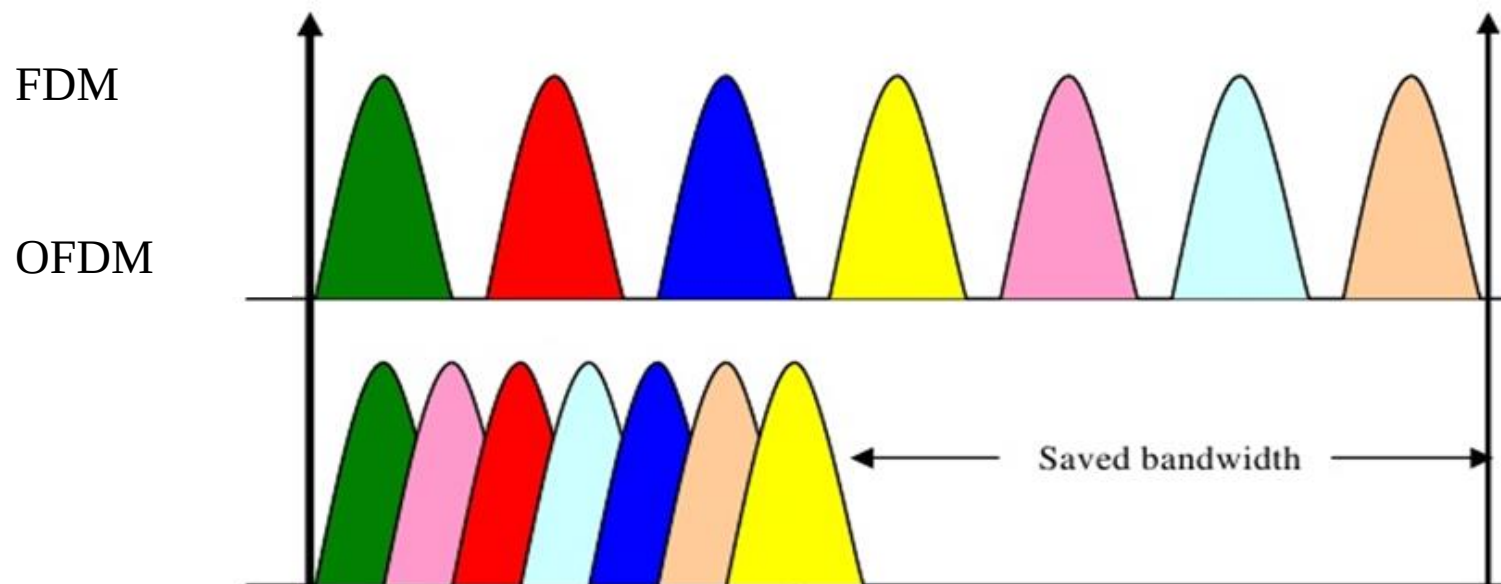


Block Diagram of FDM Communication System:



5.8.3 Orthogonal Frequency Division Multiplexing (OFDM)

- **OFDM** uses partially overlapping **orthogonal** subcarriers.
- **OFDM** has higher spectral efficiency than **Frequency Division Multiplexing (FDM)** and is applied in systems such as 5G and WiFi.



5.8.4 Time Division Multiplexing (TDM)

The theoretical basis of TDM is the **Sampling Theorem**.

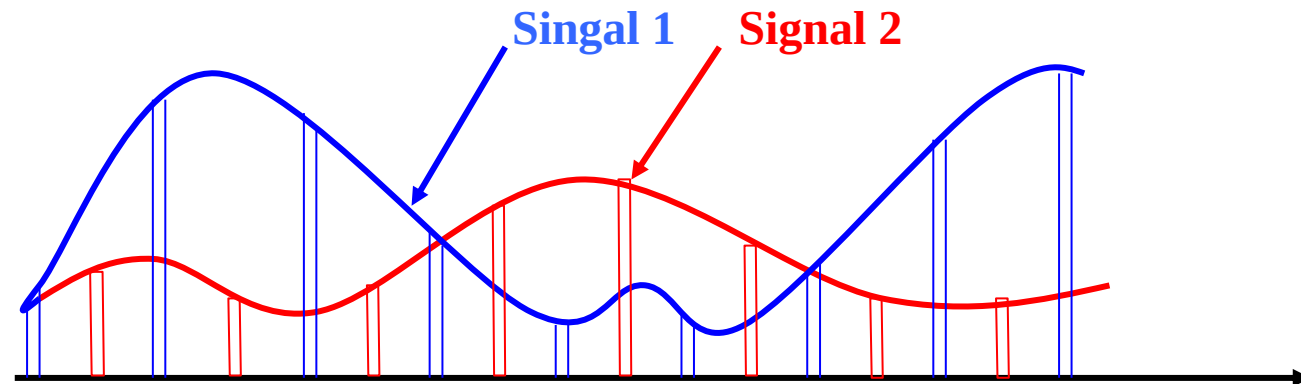
A signal in the range $-f_m \sim +f_m$ can be uniquely determined by sampling values with an interval of $1/(2 f_m)$. The original continuous signal can be correctly recovered from these instantaneous sampling values.

The channel is occupied only at sampling instants, and the remaining idle time can be used to transmit sampling signals of the second, third, ... and other channels.

TDM is realized by arranging the sampling values of each channel in an orderly manner.

At the receiving end, these sampling values are separated by an appropriate synchronous detector.

Time Division Multiplexing for Two-Channel Signals:



1. Repeat the examples in the lecture notes.
2. Given the spectrum of a signal $f(t)$ shown in Fig.1, the signal $f_1(t)$ is obtained by multiplying $f(t)$ with $\cos(100t)$ and the signal $f_2(t)$ is obtained by multiplying $f_1(t)$ with $\cos(100t)$ again. Then the signal $f_2(t)$ passes through an ideal low-pass filter with a cutoff frequency of ω_c to get the signal $r(t)$.
 - (1) Draw the spectrum $F_1(\omega)$ and $F_2(\omega)$ corresponding to the signals $f_1(t)$ and $f_2(t)$.
 - (2) If $r(t) = f(t)$, find the range of the cutoff frequency of the LPF ω_c .

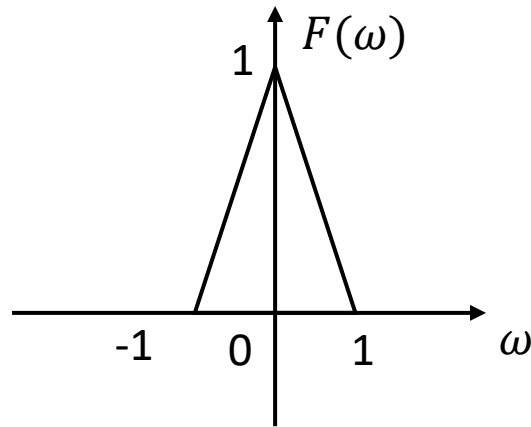


Fig.1

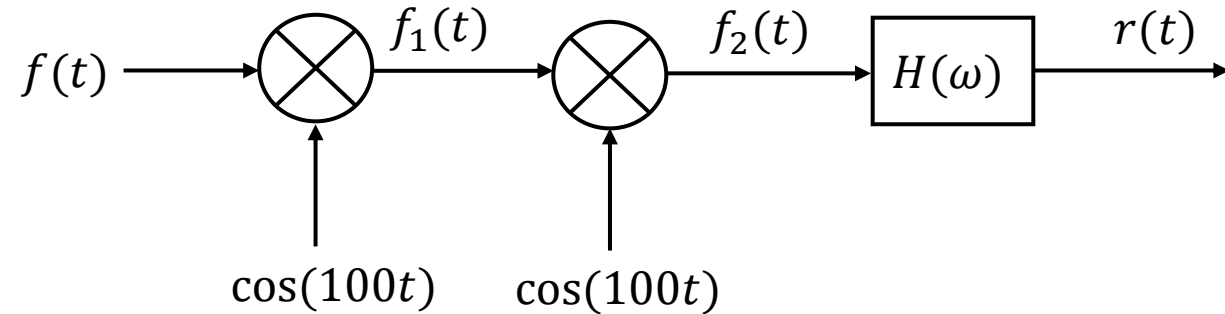


Fig.2